FOR THE PERSON WHO BOUGHT THE SENSOR

Set it up on the kitchen table first. **Then have it fitted.**

GeyserSense measures your geyser's water temperature and reports it to your home automation. This guide is the order to do things in.

The order

1. **Get it onto your WiFi**, on the kitchen table, powered from USB-C. Page 3.
2. **Test the sensor**, still on the table, with the resistor that came in the box. Page 4.
3. **Send the readings somewhere**: Home Assistant, or any MQTT broker. Page 5.
4. **Have it installed** at the geyser, by a plumber or an electrician. Page 7.

Pages 8 to 10 are for later. Page 8 is everyday use: reaching the unit, what to do when something looks wrong, and updates. Page 9 is the thermostat's cut-out, worth reading once. Page 10 is a fixed address, only if page 6 sends you there.

What you need

- The GeyserSense unit.
- The **10 kΩ resistor** supplied with it. Do not throw it away. It is how you test the unit now, and how you tell a broken sensor from a broken unit later.
- A Geyserwise `GEY013` thermostat, fitted to your geyser by a plumber or an electrician. Step 4 is written for whoever does that.
- A USB-C cable and a phone charger or a computer, to power the unit on the table.
- A phone.
- Somewhere for the readings to go. Step 3 covers that.

What's on the board

The numbers below are used throughout this guide. Orange items are the ones you touch. Dark items are for whoever installs it; the two terminals and the supply module carry mains once the unit is fitted.

The bare board, Rev A, seen from above. It is supplied like this; the enclosure is separate.

1. **Status lights** YOU
   Three lights: ERR is red, NET is green, USR is white. What each pattern means is in the table below.

2. **Button** YOU
   Hold it for ten seconds to erase every setting and return the unit to new. The white USR light blinks from five seconds as a warning. Keep holding: at ten it goes solid while the green NET light blinks fast, and the settings are gone. Let go before ten and nothing happens. A short press changes nothing.

3. **Probe terminals, GND and NTC** YOU AND INSTALLER
   The thermostat's black probe lead goes here, either way round. For the table test the 10 kΩ resistor goes here instead.

4. **Mains terminals, N and L** INSTALLER
   230 V, fed from the geyser's isolator switch. Never touch these with the unit powered.

5. **USB-C port** YOU, FOR THE TABLE TEST
   Powers the unit from any USB charger or a computer, with nothing on the mains terminals. It is also the service port for firmware and diagnostics. Nothing else needs it.

6. **WiFi antenna** INSTALLER
   The end of the module that overhangs the board edge. Keep it clear of metal when the board is mounted. Nothing to touch.

7. **Mains supply module** INSTALLER
   Makes the unit's own supply from the mains. It carries mains, and there is nothing to adjust.

8. **Fuse** INSTALLER
   Protects the unit's own supply only. It has nothing to do with the geyser's circuit.

9. **Mounting holes** INSTALLER
   Four M3 holes in a rectangle, one at each corner. The photo's angle hides the top-left one behind the supply module.

What the lights mean

LIGHT	PATTERN	MEANING
ERR red	**Off**	The sensor is reading normally.
	Slow blink, once a second	A probe or wiring fault. Most likely a loose wire at the probe terminals, or a failed probe.
	Steady on	A fault inside the unit. The probe is not the problem.
NET green	**Steady, dipping every three seconds**	On your WiFi. The dip is a heartbeat; a light that never dips means the unit has hung.
	Slow blink, once a second	Trying to join your WiFi.
	Fast blink	The unit is offering its own setup network. Step 1 starts here.
USR white	**Blinks for about a minute**	You pressed **Blink this unit's light** on the Help page. It tells you which unit you are talking to.
	Fast blink during a button hold	Five seconds into the hold. Let go now unless you mean to erase every setting.
	Solid, with NET blinking fast	Ten seconds reached. The settings are erased and the setup network is back.

STEP 1

Get it **onto your WiFi**

Do this on the kitchen table, before anything goes near the geyser. Once fitted, the unit is usually up in the roof next to the geyser, and that is no place to set it up, or to reset it later.

> **Never on mains while the board is out of its enclosure.**
> Steps 1 and 2 run on USB-C only, with the board out of its enclosure and nothing on the mains terminals **(4)**. Once mains is connected, the solder joints on the underside of the board carry 230 V; they are marked there. A bare board on mains is a shock hazard. Mains goes on only once the board is fixed in its enclosure, by whoever installs it.

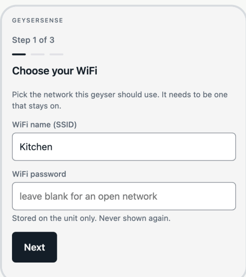
1. Power the unit from the USB-C port **(5)** with a phone charger or a computer, with nothing on the probe terminals. It creates its own WiFi network, named `gs-` followed by six characters, for example `gs-0e8770`. The NET light **(1)** blinks fast while that network is on offer. The red ERR light blinks slowly too: nothing is on the probe terminals, and that is what a probe fault looks like. Expected here.
2. On your phone, join that network. A setup page opens on its own. If it does not, open a browser and go to `192.168.4.1`.
3. Choose your home WiFi and enter its password. Pick a network that stays on, not a guest network that reboots nightly.
4. Set a **sensor password**, eight characters or more. It protects changes to the settings. It does not hide the temperature, which stays visible to anyone on your network.
5. Give the geyser a name you would recognise: "Kitchen Geyser", not "Sensor 1". Press **Connect**.

The page may pause for a few seconds while it joins. The unit has one radio, so joining your network briefly interrupts the one it made for you. If your phone does not re-join by itself, re-select the `gs-` network and the page comes back.

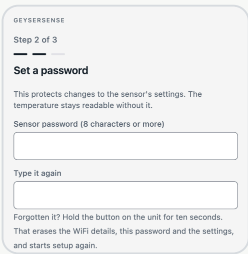
When it succeeds you are shown the unit's three **addresses**. Write down the last two: the permanent name, like `gs-0e8770.local`, which is the setup network's name and never changes; and the number, like `192.168.0.42`, for a phone that cannot open the name (some Android phones cannot). The first, like `kitchen-geyser.local`, is made from the geyser's name and changes if you rename it, so do not build on it.

Prefer the permanent name. The number is handed out by your router and can change, usually after a power cut. The name keeps working when that happens. Then take the offer to **add it to your home screen**.

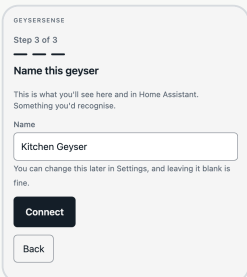
What you will see: **getting it onto your WiFi**



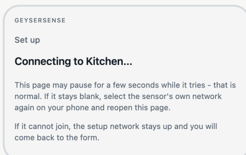
Step 1 of 3. Your home WiFi's name and password. The examples here use "Kitchen".



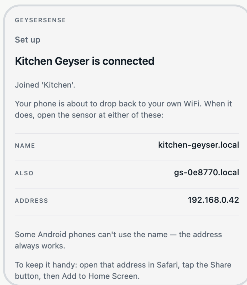
Step 2 of 3. The sensor password, eight characters or more. It protects the settings, not the temperature.



Step 3 of 3. The geyser's name, as Home Assistant will show it. Press **Connect**.



Connecting. The page may pause here. If it stays blank, re-join the `gs-` network and reopen it.



Connected. The addresses. Write down the `gs-` name and the number.

STEP 2

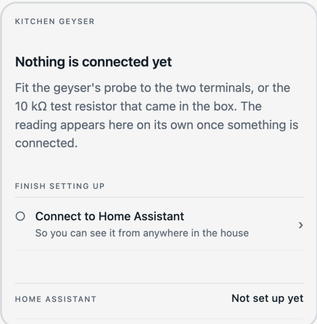
Test the sensor **on the table**

Still on USB-C, still on the table. Months from now, this is the difference between "the sensor has failed" and "the geyser has failed". The unit takes a reading every 30 seconds and wants three in a row that agree, so each change below takes up to a minute and a half to show. The pages update on their own.

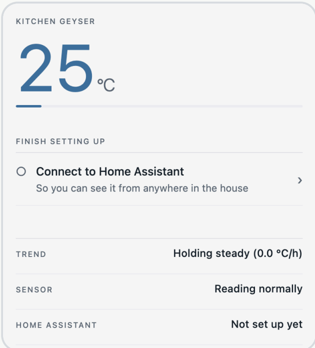
1. Open the unit's page at the address you wrote down. Home says **Nothing is connected yet** and asks for the probe or the 10 kΩ resistor.
2. Fit the resistor across the two probe terminals, marked GND and NTC **(3)**. Either way round. Home shows a reading on its own, about 25 °C, and the ERR light goes off.
3. Now open **Help › Test the sensor**. It opens on **The unit is reading correctly**. Press **See what a fault looks like.**
4. Now take it off. The page says **The fault was noticed**, and the ERR light blinks again. Press **All done**.

What you will see: **testing the sensor**

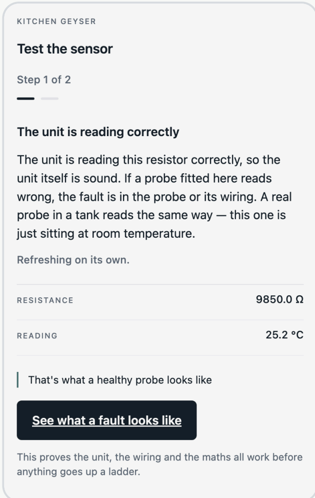
The four screens of step 2, once the unit is on your WiFi. Open the unit's page at the address you wrote down.



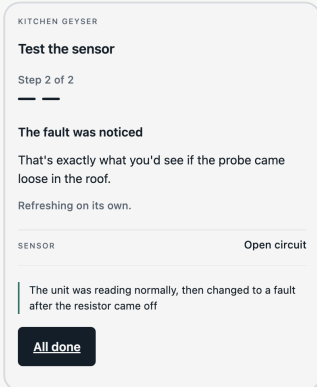
Home, before anything is fitted. Nothing is connected yet. Fit the resistor now, as Home asks.



Home, resistor fitted. The reading appears on its own, at room temperature. Now open **Help › Test the sensor**.



Step 1 of 2. The test opens on a pass: the unit is reading correctly, about 9,850 Ω. Press **See what a fault looks like.**

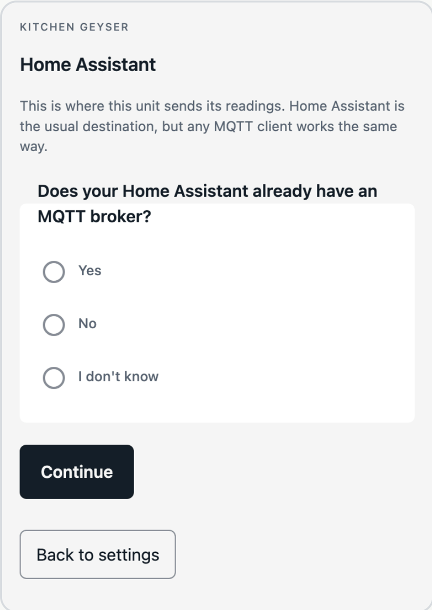


Step 2 of 2. Take the resistor off. The fault was noticed: the sensor row says Open circuit. Press **All done.**

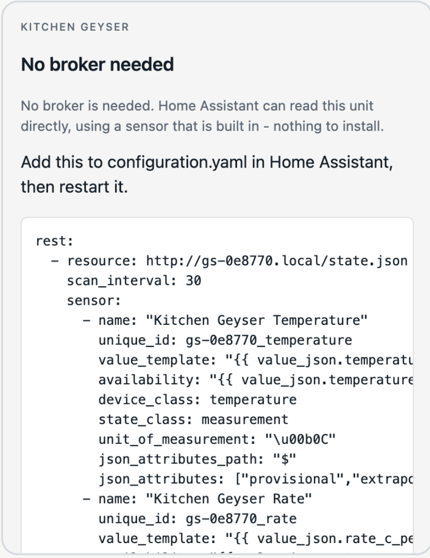
STEP 3

Send the readings **somewhere**

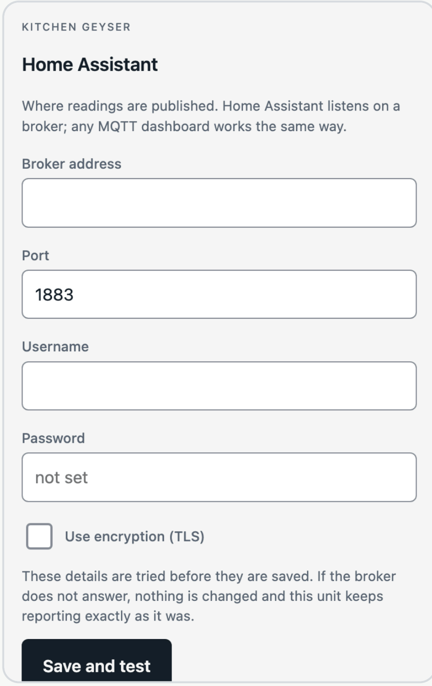
On Home, open **Connect to Home Assistant**. It is the same page as **Settings › Home Assistant**. The first time you open any settings page you are asked for the sensor password from step 1. The page asks one question: **Does your Home Assistant already have an MQTT broker?** Choose **Yes**, **No** or **I don't know**, then press **Continue**. No and I don't know go the same way, because no broker is needed for that path.



The question. Most people answer No or I don't know, and that is the simpler path.



No broker needed. The block to paste into Home Assistant, with this unit's own address already in it. The page continues below the block.



Yes: the broker form. The details are tried before they are saved, so a typo tells you immediately.

If you answered No, or I don't know

The **No broker needed** page gives you a block to paste into Home Assistant's `configuration.yaml`, with your unit's address already filled in. Copy the whole block, paste it, and restart Home Assistant. Five sensors appear within a minute: the temperature, how fast it is changing, the trend, how many times it has gone over your limit, and whether the probe has a problem. Nothing to install, no add-on, no account.

The block uses the unit's **permanent name**, not its number. That is deliberate, and the next page says why.

If no sensor appears, replace the name in the block with the unit's number. It is shown at the foot of the same page as **This unit's IP address**. Home Assistant reaches the unit by name only if your installation can resolve `.local` addresses. Most can, but not all. It depends on how Home Assistant was installed.

The five sensors arrive separately rather than grouped under one device; that is how Home Assistant reads a unit directly. Put them in an area once they appear, under **Settings › Devices & services › Entities** in Home Assistant, and your dashboard and area-based automations can find them.

STEP 3, CONTINUED: IF YOU ANSWERED NO, OR I DON'T KNOW

Why the block uses the name, and what to do if you used the number

A name follows the unit. A number is your router's to hand out, and the router can change it, most often when the whole house reconnects after a power cut. When that happens, a sensor read by its number quietly stops updating while the unit itself carries on working perfectly, which is what makes it hard to spot. That is why the block uses the permanent name: it keeps working whatever the number does.

If Home Assistant could not open the name and you used the number instead, keep that number from changing. It is not hypothetical: on our own bench the router handed the unit an address it had earlier given to another device, and Home Assistant spent the afternoon talking to the wrong one. Pick whichever is easier:

- **Reserve the address on your router.** Look for "DHCP reservation", "static lease" or "bind IP to MAC" in its settings, and tie the unit's address to it permanently. This is the better fix. It works no matter how Home Assistant is installed, and it also stops the router handing that address to something else.
- **Or set a fixed address on the unit**, under **Settings › Network › Address on your network**. Page 10 shows the screens. You will need a free address on your network and your router's address. The unit checks that a fixed address actually works before keeping it, and falls back to Automatic if it does not.

The unit also watches for this itself. If its address changes and it looks like something is reading from it, it says so on its home page and tells you what to update.

This path works on your local network only. If you want to see your geyser from outside the house, use the broker path instead.

If you answered Yes, or you want to read it from anywhere

This is the better path if you already run **Mosquitto** in Home Assistant, or if you use a hosted broker. It also sidesteps the address problem entirely. The unit sends readings *out* to the broker, so nothing needs to know where the unit lives, and a changed address costs you nothing.

To add Mosquitto in Home Assistant: **Settings › Add-ons › Add-on Store › Mosquitto broker › Install › Start**. Then **Settings › Devices & Services**, and accept the MQTT integration it offers to set up. Create a Home Assistant user for the sensor to log in as (**Settings › People › Add person**, with "Allow login" on). The broker uses your Home Assistant accounts.

Back on the unit's **Home Assistant** page: enter the **Broker address**. That is your Home Assistant machine's number, the one you open Home Assistant at, without the `http://` or the `:8123`; Home Assistant shows it under **Settings › System › Network**. Leave **Port** at 1883, enter the **Username** and **Password** you just created, and press **Save and test**. The unit tries the details before saving them. If the broker does not answer, nothing is changed and the unit keeps reporting as it was.

If it fails, the page says why: a wrong port or address gets "Nothing answered in time", a wrong login gets "The broker refused these credentials". Fix it and press **Save and test** again. When it succeeds, Home's **Home Assistant** row says **Connected to broker**, and the sensor and its companions appear in Home Assistant on their own. Nothing to configure at the Home Assistant end.

For a hosted broker, reading your geyser from another city or another country: enter the address and port your provider gives you, usually 8883, tick **Use encryption (TLS)**, and press **Save and test**. Encryption is not optional over the internet.

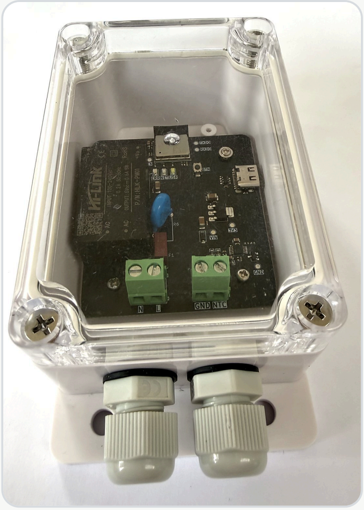
STEP 4

Have it **installed**

Once the table test passes and readings are arriving, have the unit and the thermostat fitted. **We recommend a registered electrician or a plumber.** Many owners replace a thermostat themselves. This page is written so that it is clear either way, and it is the page to hand to whoever does the work.

For the installer

- **Isolate the geyser first.** Switch off at the isolator next to the geyser and at the distribution board, and confirm the circuit is dead before touching anything.
- **The thermostat.** The existing thermostat comes out and the Geyserwise `GEY013` goes in its place. Its mains wiring is the same as the standard manually-set thermostat it replaces. The difference is the extra black lead, which is the probe this unit reads.
- **The probe lead.** Cut the black connector off the probe lead. Strip the two wires and screw them into the probe terminals marked GND and NTC **(3)**. Either way round; the probe has no polarity.
- **Power for the unit.** Take the unit's supply from the load side of the isolator switch next to the geyser, into the terminals marked N and L **(4)**. The isolator then cuts the unit together with the geyser. The unit is not wired into the element circuit anywhere.
- **The enclosure.** One cable gland for the mains cable, the other for the probe lead. Fix the board on its four mounting holes **(9)**, with the WiFi antenna **(6)** clear of metal. Nothing else on the board is touched.



Fitted in an IP66 enclosure, two glands: mains in one, the probe lead in the other.

> **Before the installer leaves**
>
> Open the unit's page once. You should see a plausible tank temperature and the sensor reading normally. If the temperature looks wrong, or the sensor shows a fault, resolve it while the installer is still there. The 10 kΩ resistor is the quickest way to tell which half is at fault: with it in place of the probe lead, a healthy unit reads normally, and the fault is in the probe or its wiring. A cold tank reads below the range the probe was measured over, and the unit says so under the temperature. That is not a fault; page 8 explains it.

LIVING WITH IT

When something looks wrong

The unit's own pages will tell you more than this guide can, because they can see the actual state. Start with the home page. A fault is stated there in plain language, with the most likely cause and a link to what to check.

Three things worth knowing in advance:

- **A sensor fault does not mean your geyser has stopped working.** The unit only measures. If the sensor fails, your geyser heats exactly as before. You have just lost visibility.
- **A note under the temperature saying it is outside the measured range is not a fault.** The probe's conversion was measured over the hot part of a geyser's range. A tank that has been standing, or a unit on the table before it is fitted, reads outside it. The unit says the same on its home page: the reading is still shown, and it should be treated as approximate until the water is hot. It settles by itself as the geyser heats. Nothing to do.
- **The water has gone cold and the sensor looks fine.** The unit reads the water, not the thermostat. The next page says what it can tell you, and what it cannot.

If you need help, open **Help › Technical details** and tap **Copy diagnostics**, then paste it into your message. It needs no password. It carries everything needed to tell a healthy unit from a faulty one without shipping it anywhere: the sensor readings, the calibration in use, the network, and whether the unit is actually talking to your broker.

If copy and paste gives you trouble, send a screenshot of the Technical details page instead. It shows less than the copied text, but it is enough to start with.

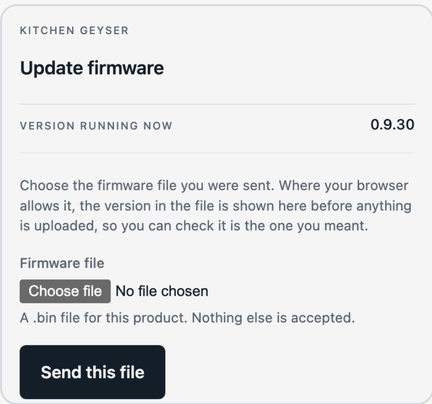
Reaching the unit later

Use the home screen shortcut from step 1. If you did not make one, use either address you wrote down. **To add one now:** open the unit's page. On iPhone tap **Share › Add to Home Screen**. On Android tap the browser menu, then **Add to Home screen** or **Install**.

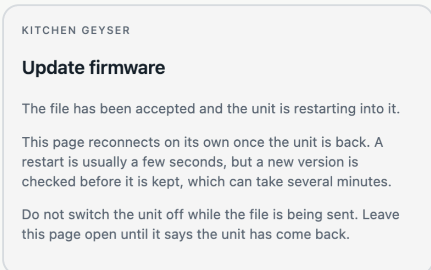
You will not need it often. Once the readings are flowing, Home Assistant is where you look. This page is for changing something or working out what is wrong.

Keeping it **up to date**

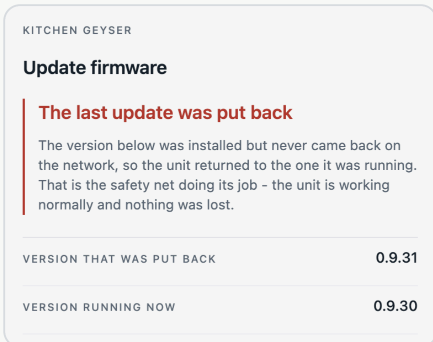
Now and then there is a new version of the firmware, sent to you as a `.bin` file. A working unit keeps working without it, so install it when it suits you. Open **Settings › Update firmware**; you are asked for the sensor password from step 1. Choose the file, check the version the page reads out of it, press **Send this file** and leave the page open. The unit restarts into the new version and the page comes back on its own, saying **Update complete**. If the new version does not come back on the network, the unit puts the old one back by itself and says so. Nothing is lost either way, and nothing needs anyone to reach the unit in the roof.



Update firmware. The version running now, and the file field. Choose the file you were sent, then **Send this file**.



While it restarts. Leave the page open; it reconnects by itself. A new version is checked before it is kept, which can take several minutes.



The last update was put back. The new version never came back on the network, so the unit returned to the one it was running. Both are named.

LIVING WITH IT, CONTINUED

The cut-out marked 90 °C, **read this once**

The `GEY013` thermostat contains a **mechanical thermal cut-out, marked 90 °C** on the part. If the water gets that hot, it cuts the element circuit. It stays cut until someone presses the reset button on the thermostat **by hand**.

That marked 90 °C is a printed figure, and no tolerance is published with it. A mechanical cut-out trips in a band rather than at a number, and the sensing element sits inside a stem that lags the water around it. So yours may trip a little under the marked figure or a little over. **Treat 90 as approximate.**

This sensor cannot see whether the cut-out has tripped. It reads water temperature and nothing else. So the symptom you will see is a geyser that has simply stopped heating, with the sensor reporting perfectly normal readings the whole time.

What it can do is remember the over-heating that trips it. The thing that trips a cut-out is water getting too hot. By the time you find the cold shower, the water has cooled and taken the evidence with it. The unit watches for that and writes it down: the hottest the water has ever got, when, and how many times it went over your limit. That record survives a power cut. It is cleared only when you clear it.

So if the water went over the limit and you have not seen it yet, **the unit tells you on its home page**: the temperature it reached and when. That is your first clue, and it is usually the whole answer. Water that hot means something ran the element on longer than it should have, and the cut-out very likely did its job. Tap **Got it** once you have dealt with it. The notice stays away until it happens again.

The limit the record is kept against is yours to set, on **Settings › Temperature record**, between **60 °C and 85 °C**. It defaults to 85 °C, which is also the highest you may set it: deliberately below the trip on any unit rather than merely below the marked figure. Set it there and the record catches the water approaching the cut-out. A limit up at the cut-out itself might never be reached, because the cut-out kills the element first and the water stops climbing.

It is still not a cut-out detector, and it never claims to be. It does not detect a tripped cut-out. It tells you the water got hot. Whether the cut-out actually tripped is something you confirm by looking at it.

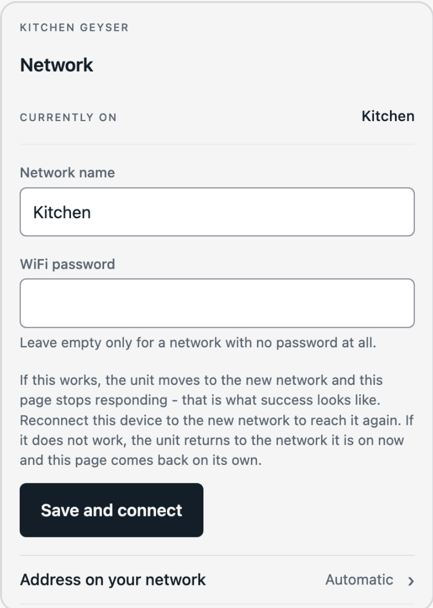
If your water goes cold and the sensor looks healthy with **no over-temperature recorded**, the cause is somewhere else: a tripped breaker, a failed element, or an automation that did not run. Check the cut-out anyway. It costs nothing. Then work through those.

The unit's **Help** page says the same thing, because the person reading this guide and the person staring at a cold shower are rarely the same person on the same day.

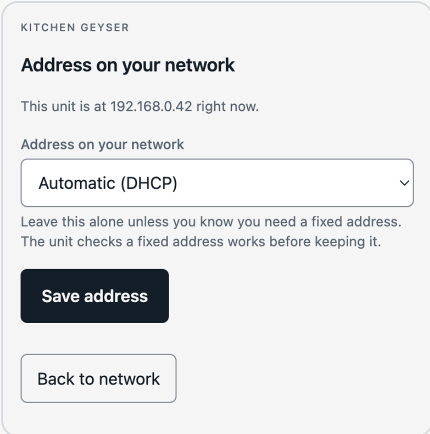
ONLY IF PAGE 6 SENT YOU HERE

Optional: **a fixed address**

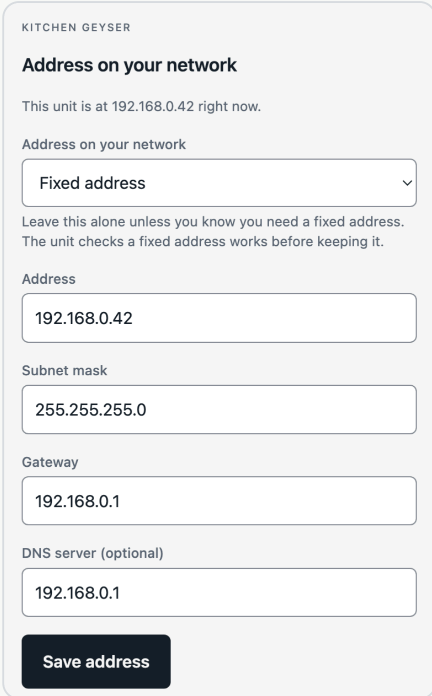
Most owners never need this. It matters only if Home Assistant reads the unit by its number and your router cannot reserve one; page 6 explains why. Open **Settings › Network**, then **Address on your network** at the foot of the page. Choose **Fixed address**, fill in the four fields, and press **Save address**. The unit tries the address before keeping it. If the address does not work, it goes back to Automatic on its own, so a mistake cannot lose the unit.



Settings › Network. The network the unit is on, the form to change it, and the address row at the foot.



Address on your network. Automatic (DHCP) is the default, and the right choice for most homes.



Fixed address. The four fields. Your router's settings page gives you the mask, gateway and DNS values.